

QRCode Analytics — Project Documentation

Visual QR code generator with dynamic links, real-time analytics, and scan tracking

Node.js • Express • QRCode Library • Vanilla JS

PROJECT OVERVIEW

A complete QR code generator with analytics dashboard. Features 7 QR types, custom styling, dynamic links, and real-time scan tracking with device/browser/OS detection. Built from scratch with zero external charting libraries.

7

QR TYPES

3

TABS

1

CLICK EXPORT

0

DEPENDENCIES

KEY FEATURES

- ✓ 7 QR types: URL, Text, Email, Phone, SMS, WiFi, vCard
- ✓ Custom colors, size, and logo upload
- ✓ Dynamic links — change destination anytime
- ✓ PNG & SVG export at custom resolution
- ✓ Real-time scan analytics dashboard
- ✓ Device, browser, OS breakdown
- ✓ 7-day trend charts
- ✓ Save/load/manage multiple QR codes
- ✓ Dynamic links — update destination anytime
- ✓ Clean, production-ready code

TECH STACK

BACKEND

Node.js + Express

FRONTEND

Vanilla JavaScript

QR LIBRARY

QRCode.js

STORAGE

JSON (file-based)

CHARTS

Canvas + CSS

ICONS

Font Awesome 6

QR TYPES SUPPORTED

- ✓ **URL** — Website links, landing pages
 - ✓ **Text** — Plain text, codes, messages
 - ✓ **Email** — mailto: with subject/body
 - ✓ **Phone** — tel: links for click-to-call
- ✓ **SMS** — Pre-filled text messages
 - ✓ **WiFi** — SSID + password for auto-connect
 - ✓ **vCard** — Contact cards with name, org, phone, email

API ENDPOINTS

GET	/api/qrs	List all QR codes
POST	/api/qrs	Create new QR code
PUT	/api/qrs/:id	Update QR code
DELETE	/api/qrs/:id	Delete QR code
GET/POST	/api/qrs/:id/generate	Generate PNG/SVG image
GET	/scan/:id	Track scan & redirect
GET	/api/qrs/:id/analytics	Get scan analytics